Physics and Imaging in Radiation Oncology 40 (2026) 101049

Contents lists available at ScienceDirect

Physics and Imaging in Radiation Oncology

journal homepage: www.sciencedirect.com/journal/physics-and-imaging-in-radiation-oncology

Original Research Article

A lightweight deep learning model for automated segmentation of Gynecologic organs and cervical Tumors on T2-weighted magnetic resonance imaging

Awj Twam [a,*], Megan C. Jacobsen [a], Adrian E. Celaya [a], Rachel Glenn [a], Peng Wei [d], Jia Sun [d], Lilie Lin [b], Ann Klopp [b], Aradhana M. Venkatesan [c], David Fuentes [a]

[a] Department of Imaging Physics, The University of Texas MD Anderson Cancer Center, Houston, TX, USA
[b] Department of Radiation Oncology, The University of Texas MD Anderson Cancer Center, Houston, TX, USA
[c] Department of Abdominal Imaging, The University of Texas MD Anderson Cancer Center, Houston, TX, USA
[d] Department of Biostatistics, The University of Texas MD Anderson Cancer Center, Houston, TX, USA

ARTICLE INFO

Keywords:
Cervical cancer
Tumor Segmentation
T2-weighted MRI
Deep learning

ABSTRACT

Background and Purpose: Manual segmentation of gynecologic organs and cervical tumors for radiotherapy planning is time-consuming and variable. Automated segmentation on routine T2-weighted magnetic resonance imaging (MRI) remains limited. The aim of this study was to evaluate a lightweight deep learning model for automated segmentation of gynecologic organs and cervical tumors on T2-weighted MRI.

Materials and Methods: This work applied a two-stage lightweight deep learning model (PocketNet) to segment the cervix, vagina, uterus, and tumor(s) on T2-weighted MRI in 102 patients with cervical cancer undergoing definitive radiotherapy. Model performance was assessed using the Dice-Sorensen coefficient (DSC) and 95th percentile Hausdorff distance (Haus95) on internal data and validated on an external dataset. A full nnU-Net model trained on the same internal dataset served as a benchmark for segmentation accuracy and computational efficiency.

Results: On the institutional dataset, PocketNet achieved mean DSC values exceeding 70% for tumor segmentation and 80% for organ segmentation. External validation on The Cancer Imaging Archive (TCIA) Cervical Cancer Tumor Heterogeneity (CCTH) collection demonstrated the model's robustness, achieving DSC scores of 67.3% for tumor segmentation and 80.8% for organ segmentation. Compared to the PocketNet architecture, nnUNet achieved similar accuracy but required approximately twice the training time, more than 35 times as many parameters, and 40 times more memory for model storage.

Conclusion: The PocketNet architecture provides reliable automated segmentation of gynecologic organs and cervical tumors on T2-weighted MRI, with performance comparable to a full-sized nnUNet while requiring substantially less memory and training time, supporting its potential integration into time-sensitive radiotherapy workflows.

1. Introduction

Cervical cancer remains a major global health burden, with over 660,000 new cases and nearly 350,000 deaths reported worldwide in 2022 [1]. Despite advances in prevention through human papillomavirus vaccination and screening, incidence and mortality rates have shown limited improvement over the past two decades [2,3]. For patients with locally advanced cervical cancer, definitive treatment consists of concurrent chemoradiotherapy followed by brachytherapy, which relies on accurate delineation of the cervical tumor and surrounding gynecologic anatomy for treatment planning.

Due to its superior soft tissue contrast, magnetic resonance imaging (MRI) is the primary imaging modality used in cervical cancer for staging and radiotherapy planning [4]. However, manual segmentation of the cervix, uterus, vagina, and tumor is time-consuming and subject to inter- and intra-observer variability, which can delay treatment initiation and limit the opportunity to perform adaptive radiotherapy [5]. Image-guided adaptive brachytherapy, which relies on accurate tumor

* Corresponding author.
E-mail address: atwam@mdanderson.org (A. Twam).

https://doi.org/10.1016/j.phro.2026.101049
Received 9 February 2026; Received in revised form 21 July 2026; Accepted 22 July 2026
Available online 24 July 2026
2405-6316/© 2026 Published by Elsevier B.V. on behalf of European Society of Radiotherapy & Oncology. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

delineation at each treatment fraction, has become the standard of care for locally advanced cervical cancer and is associated with improved local control and reduced treatment toxicity [6,7]. Automated segmentation methods have the potential to address these limitations by improving efficiency and consistency of radiotherapy workflows.

Deep learning-based segmentation methods, particularly convolutional neural networks (CNNs), have demonstrated promising performance in medical image analysis, including cervical cancer imaging. Prior studies reported high accuracy (90.8%) for detecting the presence of cervical cancer and for segmenting tumors using multi-parametric MRI [8]. Deep learning models have also been applied to auto-segmentation of clinical target volumes for cervical cancer radiotherapy planning, with several studies reporting DSC values exceeding 0.80 for organ-level segmentation [9,10]. However, many state-of-the-art segmentation models, such as full-scale U-Net-based architectures including nnU-Net [11], are computationally intensive, requiring substantial memory and training time, which may limit their deployment in resource-constrained or time-sensitive clinical environments.

To address this, a PocketNet approach was applied to segment the uterus, vagina, cervix and tumor(s). The PocketNet architecture builds on the U-Net backbone but dramatically reduces the number of parameters, minimizing the computational burden while producing similar performance to that of full-sized networks [12]. This work evaluated the performance of a two-stage PocketNet approach for image segmentation of cervical cancers on T2-weighted MRIs in patients treated with definitive radiotherapy. To validate model robustness, a publicly available dataset from The Cancer Imaging Archive (TCIA) was utilized.

2. Materials and methods

All data used in this study were collected and analyzed under an approved Institutional Review Board protocol (MD Anderson IRB Protocol # 2025–0861, Imaging Biomarkers and Clinical Outcomes in Gynecologic Tumors: A Retrospective Analysis of Borderline and Malignant Cases).

A total of 232 pre-treatment T2-weighted MR images were collected from 102 patients diagnosed with cervical cancer who underwent definitive chemoradiotherapy. Some patients had more than one MRI scan in the dataset, as repeat imaging was performed during the course of radiotherapy to facilitate brachytherapy applicator placement and adaptive treatment planning; only the baseline pre-treatment scans were used for model training and evaluation, with duplicate scans removed. Two volumetric contours were manually segmented per patient: 1) the combined uterus, cervix, vagina, and tumor, collectively labeled as ‘organ’ and 2) the cervical tumor alone. Manual segmentations were performed by trained staff and subsequently reviewed and approved by a fellowship-trained abdominal radiologist with 22 years of clinical experience, serving as the ground truth for model training and evaluation. External validation was performed using the Cervical Cancer Tumor Heterogeneity (CCTH) collection, a publicly available dataset from TCIA comprising 23 subjects with cervical cancer staged according to the International Federation of Gynecology and Obstetrics (FIGO) classification system (stages IB2-IVA) with standardized multi-parametric imaging and gold-standard tumor contours on T2-weighted MRI [13]. Acquisition details for both datasets are provided in the Supplementary Material (Table S1).

Inclusion criteria for MR images used for training, validation, and testing were as follows: 1) complete visualization of the cervix, vagina, uterus, and tumor; 2) presence of entire organ segments within the imaging volume; 3) acceptable image quality, defined as cases where boundaries of the organ and tumor were visually identifiable without the aid of a pre-existing contour. Quality assessment was performed by visual review prior to inclusion. Cases exhibiting severe motion artifacts, signal dropout, or other imaging failures that precluded reliable boundary identification were excluded. Of the 23 CCTH subjects, one was excluded due to the absence of a T2-weighted sequence and four were excluded due to failures in resampling to a consistent orientation, yielding 18 cases for external validation.

All images were prepared using the preprocessing pipeline from the Medical Imaging Segmentation Toolkit (MIST), an integrated framework within which PocketNet operates. The toolkit processed files in Neuroimaging Informatics Technology Initiative (NIfTI) format and output reoriented, resampled, windowed, and normalized NumPy (Numerical Python) arrays. To maintain consistency with the model's training data, images acquired from TCIA, initially in Digital Imaging and Communications in Medicine (DICOM) format, underwent conversion to NIfTI. Each image was then resampled and reoriented from the sagittal view to an axial view in the right-anterior-inferior orientation, to standardize orientation and spacing across all datasets. Image volumes were resampled using label-preserving interpolation. Intensity normalization was performed using Z-score normalization, where voxel intensities within each image were standardized to zero mean and unit variance based on non-zero voxels. All MR images were prepared using a patch size of $256 \times 256 \times 32$ and a pixel spacing of $0.47 \times 0.47 \times 5$ mm. The patch size was derived from the median resampled image size by selecting the nearest power of two less than or equal to each dimension, constrained by a maximum patch size.

As depicted in Fig. 1, the network was constructed from a composition of convolution and down sampling operations that extracted features along a contracting path. An expanding path consisted of convolution and up sampling operations with ‘long skip’ connections to integrate features from the corresponding down sampling operations. Four resolution levels were used. At a given resolution, the feature maps of all preceding layers were used as inputs, and its own feature maps were used as inputs into all subsequent layers. The key difference in the PocketNet architecture is that the number of layers was not heuristically increased at each resolution, reducing the total parameter count relative to standard UNet-based architectures. Each convolutional operation used a 3×3 kernel size followed by a batch normalization and a rectified linear unit (ReLU) activation function.

An Nvidia Quadro RTX 8000 graphical processing unit (GPU) was used with a batch size of 4 and 1000 epochs. Network weights were optimized using the Adaptive Moment Estimation (ADAM) optimizer with a learning rate of 0.0001 and a cosine annealing scheduler. Automatic mixed precision was used during training to reduce the time and memory requirements. Classification tasks included Dice with categorical cross-entropy as the loss function. Regularization features included deep supervision and segmentation tasks using the L_2 norm. In addition, residual blocks were utilized in the architecture to improve training by resolving the degradation problem with the use of skip-connections around convolution layers. Postprocessing was applied by retaining the largest connected component of each predicted mask, eliminating noise and small islands distant to the body of the contour. Segmentation accuracy was evaluated using the Dice-Sorensen coefficient (DSC) and 95th percentile Hausdorff distance (Haus95) as primary outcome metrics. The performance of PocketNet with default parameter settings was also assessed separately for comparison with customized configurations.

2.1. Experimental design

A two-stage segmentation approach was performed using PocketNet to evaluate the architecture's performance for T2-weighted MRI-based automated segmentation of gynecologic organs and cervical tumors. A two-stage approach was chosen because cervical tumors are anatomically contained within the gynecologic organ complex. Segmenting the organ in the first stage and constraining tumor segmentation to within that boundary in the second stage reduces the effective search space for tumor delineation and false positive predictions in distant pelvic structures that may exhibit similar T2-weighted signal characteristics to tumor tissue.

In the first stage, a five-fold cross-validation was performed on the initial training dataset of 35 patients to generate an organ segmentation

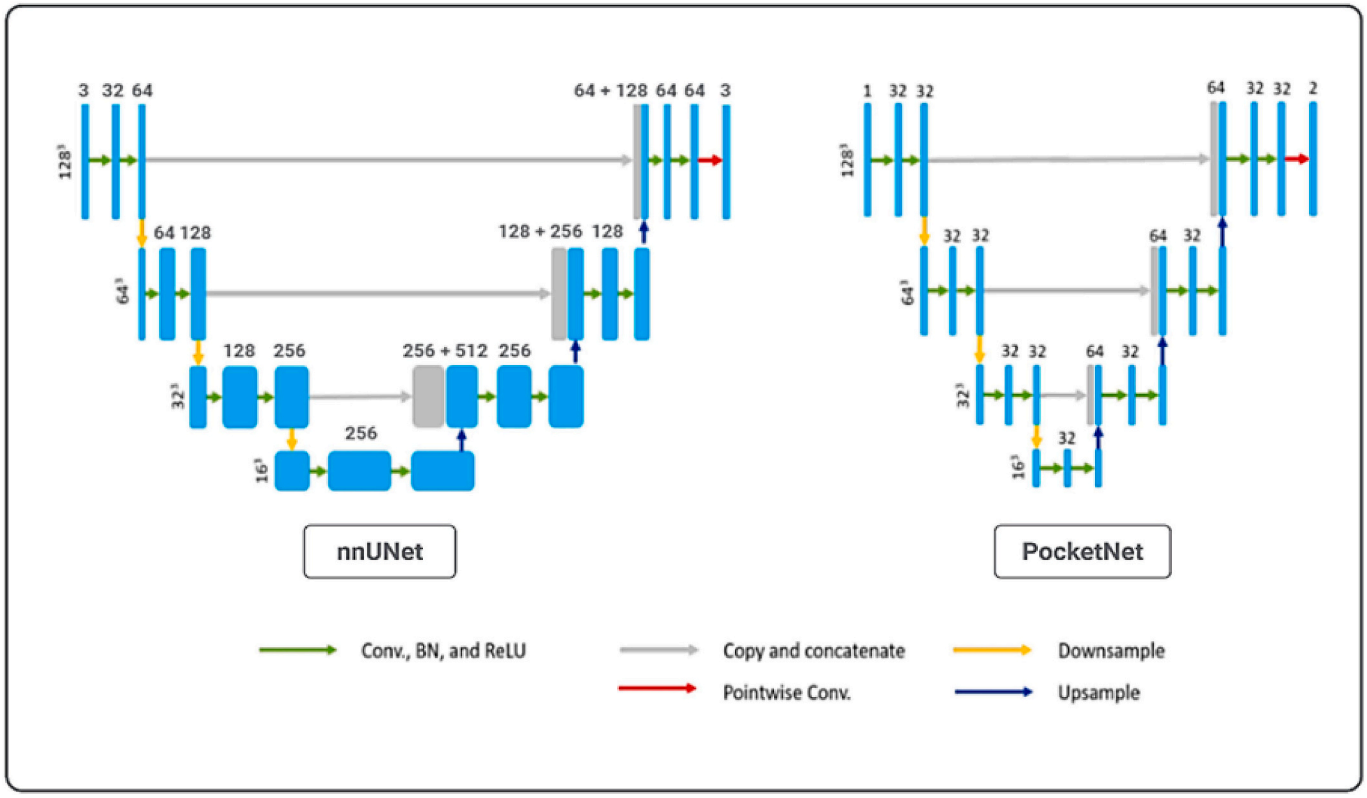


Fig. 1. Architectural comparison between nnUNet and PocketNet. The nnUNet framework (left), follows a U-Net-style encoder-decoder architecture with increasing feature depth at lower spatial resolutions. In nnUNet, convolutional blocks employ instance normalization and ReLU activation.

model. Inference was then performed to generate organ prediction masks on 102 patients. In the second stage, the organ and tumor masks for the 102 patients were combined into a single dataset. A further five-fold cross validation was performed on this dataset, incorporating the T2-weighted image alongside the combined organ-and-tumor mask. Patient-level splitting was applied across all folds to prevent data leakage.

A comparative analysis was conducted between PocketNet and a full-sized nnUNet architecture, each trained on the same dataset using identical training protocols. Both models were trained using batch sizes of 1 and 4 with higher batch sizes excluded due to GPU memory constraints with the full model. Training memory usage was estimated using double precision, accounting for 8 bytes per parameter.

To evaluate model generalizability, the PocketNet model trained on the institutional dataset (102 cases) was applied to the 18 CCTH T2-weighted MRIs, generating automated segmentations for both gynecologic organs and cervical tumors.

3. Results

On the MD Anderson Cancer Center (MDACC) dataset, mean DSC was 0.81 ± 0.12 for gynecologic organ segmentation and 0.71 ± 0.21 for tumor segmentation. Corresponding mean Haus95 values were 64.2 ± 65.9 mm for organs and 38.6 ± 56.3 mm for tumors. On the external dataset, mean DSC was 0.81 ± 0.11 for organs and 0.67 ± 0.23 for tumors, with mean Haus95 values of 14.8 ± 7.5 mm for organs and 16.0 ± 13.1 mm for tumors (Fig. 2, Table 1).

A quantitative comparison between PocketNet and a full-sized

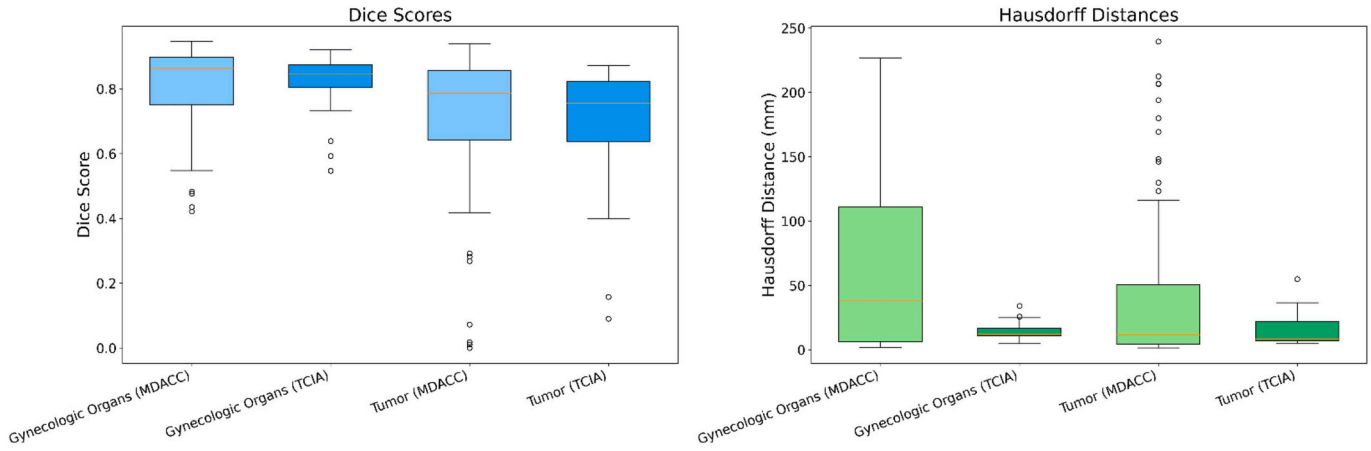


Fig. 2. Boxplots of Dice Sorensen Coefficient and Hausdorff Distances for Gynecologic Organs and Tumor.

Table 1
Computational and Segmentation Comparison Between PocketNet and nnUNet.

Model	Batch Size	Train Time (s)	Model Size (MB)	# Params (M)	Cervix Dice	Tumor Dice	Cervix Haus95 (mm)	Tumor Haus95 (mm)
PocketNet	4	696.6	0.57	0.15	0.81 ± 0.12	0.71 ± 0.21	64.2 ± 65.9	38.6 ± 56.3
nnUNet	4	1362.8	20.9	5.5	0.80 ± 0.13	0.68 ± 0.25	56.7 ± 67.0	33.7 ± 54.3

nnUNet model was performed using identical training data, preprocessing, and cross-validation folds (Table 1).

PocketNet required 696.6 s of training time, had a model size of 0.57 MB, and contained 0.15 million parameters. In comparison, nnUNet required 1362.8 s of training time, had a model size of 20.9 MB, and contained 5.5 million parameters.

PocketNet achieved a mean DSC of 0.81 ± 0.12 for cervix segmentation and 0.71 ± 0.21 for tumor segmentation. Corresponding nnUNet DSC values were 0.80 ± 0.13 for cervix segmentation and 0.68 ± 0.25 for tumor segmentation. Mean Haus95 values were 64.2 ± 65.9 mm (cervix) and 38.6 ± 56.3 mm (tumor) for PocketNet, compared with 56.7 ± 67.0 mm and 33.7 ± 54.3 mm for nnUNet, respectively. No statistically significant differences were observed between models for any metric: organ DSC ($t = 0.77$, $p = 0.44$), tumor DSC ($t = 1.22$, $p = 0.22$), organ Haus95 ($t = 0.87$, $p = 0.39$), and tumor Haus95 ($t = 0.71$, $p = 0.48$), based on paired *t*-tests and confirmed by Wilcoxon signed-rank analysis (all $p > 0.05$).

Segmentation accuracy was influenced by tumor size, anatomical location, and image quality. Very small tumors difficult to visualize on T2-weighted yielded the lowest DSC values, while tumors with standard MRI presentation achieved high performance. Very large tumors with adnexal invasion showed more variable and reduced performance.

4. Discussion

This study evaluated a lightweight two-stage deep learning model for automated segmentation of gynecologic organs and cervical tumors on T2-weighted MRI in patients with cervical cancer undergoing definitive radiotherapy. The model achieved mean DSC of 0.81 for organ segmentation and 0.71 for tumor segmentation on the MDACC dataset, with comparable performance on external validation (0.81 and 0.67, respectively). Segmentation accuracy was statistically equivalent to a full-sized nnU-Net across all metrics (all $p > 0.05$), while requiring 40 times less memory and half the training time, supporting its potential as a computationally efficient tool for radiotherapy planning. Prior work demonstrated that deep learning (DL) based autosegmentation reduced contouring time by approximately 30 min, with 84% of physicians reporting clinical satisfaction, supporting its growing adoption in routine workflows [14].

The PocketNet paradigm introduces several optimizations well-suited for resource-constrained environments while maintaining high segmentation performance. Prior benchmarking compared PocketNet against full-sized U-Net, ResNet, DenseNet, and HRNet variants, reporting inference throughput improvements of 1–17% depending on image size [12]. At 128 × 128 × 128 input resolution, PocketNet achieved 8.33 images/s compared with 7.84 images/s for U-Net (5.9% faster), with similar advantages observed across other architectures. These results demonstrated that PocketNet achieved modest but consistent acceleration in inference, while its major benefit remains reduced memory usage and faster training cycles for every batch size [12].

The performance of automatic segmentation algorithms in cervical cancer treatment presented both opportunities and challenges, as evidenced by the varying results highlighted in Figs. 3 and 4. A key observation from these results was the apparent relationship between tumor size and segmentation accuracy. Larger tumors, as seen in Fig. 3, tended to be segmented more accurately, resulting in higher Dice scores, while smaller tumors, like those in Fig. 4, often yielded lower Dice scores. Smaller tumors (< 3 cm) may be entirely confined to the cervix or lower uterine segment with less signal difference between the normal tissue and tumor on T2-weighted imaging. In contrast, larger masses were more likely to involve the full thickness of the cervix or have

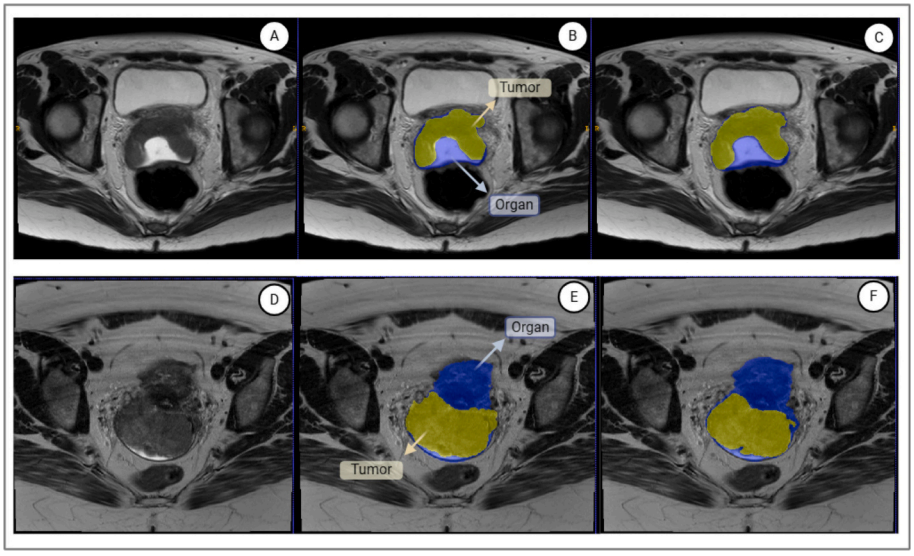


Fig. 3. Manual and automated segmentation in patients. A) Axial T2-weighted image of patient with FIGO stage IVA squamous cell carcinoma of the cervix. D) Axial T2-weighted image of patient with FIGO stage IIB adenocarcinoma of the cervix. B, E) Manual segmentation of organ (blue) and tumor (yellow). Note that the gynecologic organ contour is inclusive of the tumor to enable the two-stage segmentation model. (C, F) Automated segmentation of organ (blue) and tumor (yellow). The manual and automated contours demonstrate strong agreement with tumor DSC of 0.88 and 0.89 and organ DSC of 0.92 and 0.95, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

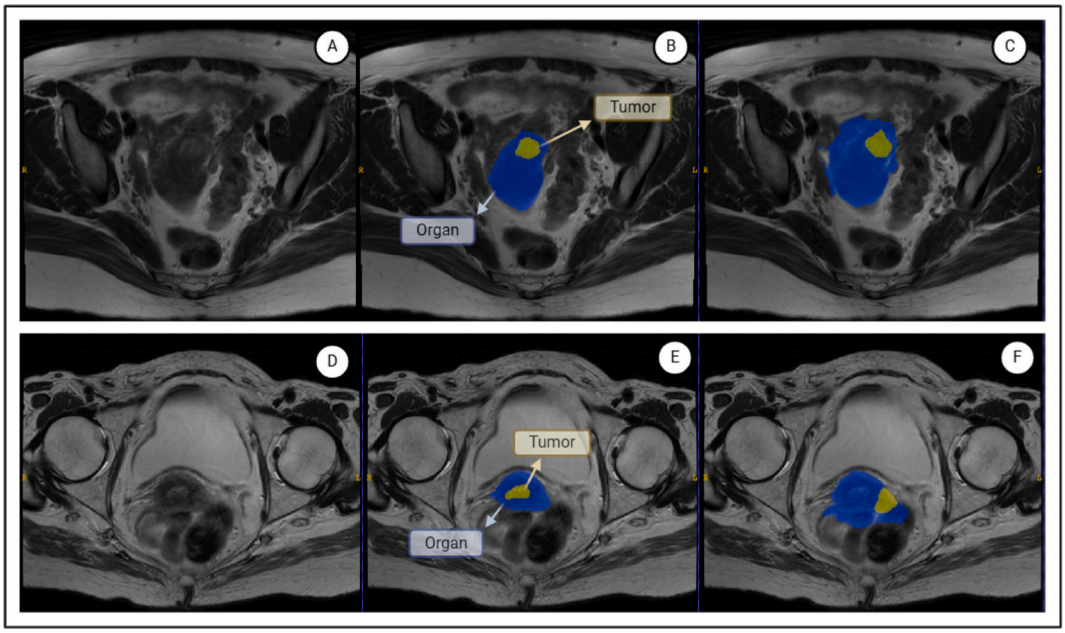


Fig. 4. Results of segmentation in two patients with FIGO Stage IIIC1 cervical cancer. A, D) Axial T2-weighted image. B, E) Manual segmentation of organ (blue) and tumor (yellow). C, F) Autosegmentation of organ (blue) and tumor (yellow). The segmentations demonstrate poor agreement in the inferior aspect of the tumor and organ, confounded by the proximity between the cervix and the adjacent rectosigmoid colon and small tumor size, with DSC values of 0.79 (C) and 0.55 (F) for organ segmentation and 0.53 (C) and 0.02 (F) for tumor segmentation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

exophytic components that were relatively simple to differentiate from the surrounding normal tissue. These findings underscore the need for further refinement and validation of these models, such as the inclusion of diffusion-weighted imaging (DWI) to identify regions of restricted diffusion that are specific to cervical tumors (Fig. 5).

Recent advancements in deep learning have brought about significant improvements in the automatic segmentation of medical images, including that of cervical cancer. In a recent study conducted by Kano et al., a deep learning model applied to DWI achieved a mean DSC of 0.77 for cervical tumor segmentation [15]. This result demonstrated the robustness of the PocketNet model as compared to existing DL methods. The comparable performance of the present T2-weighted-based model, which achieved a mean DSC of 0.71, to this DWI-based model suggested that PocketNet has the potential to achieve high accuracy across various imaging sequences with further development. Another study comparing the segmentation performance of a DL algorithm (specifically, a 3D U-Net) with manual segmentations provided by two radiologists (R1 and R2) reported median DSCs of 0.60 for R1 and 0.58 for R2 [16]. In addition, a study utilizing a U-Net based triple-channel model using sagittal diffusion-weighted MRI (DWI) with b-values of 0 and 1000 s/mm^2 and apparent diffusion coefficient (ADC) maps produced a promising average DSC of 0.82, further demonstrating the capability of DL models to enable accurate segmentation of cervical cancer [17]. One of the most promising applications of DL models in cervical cancer

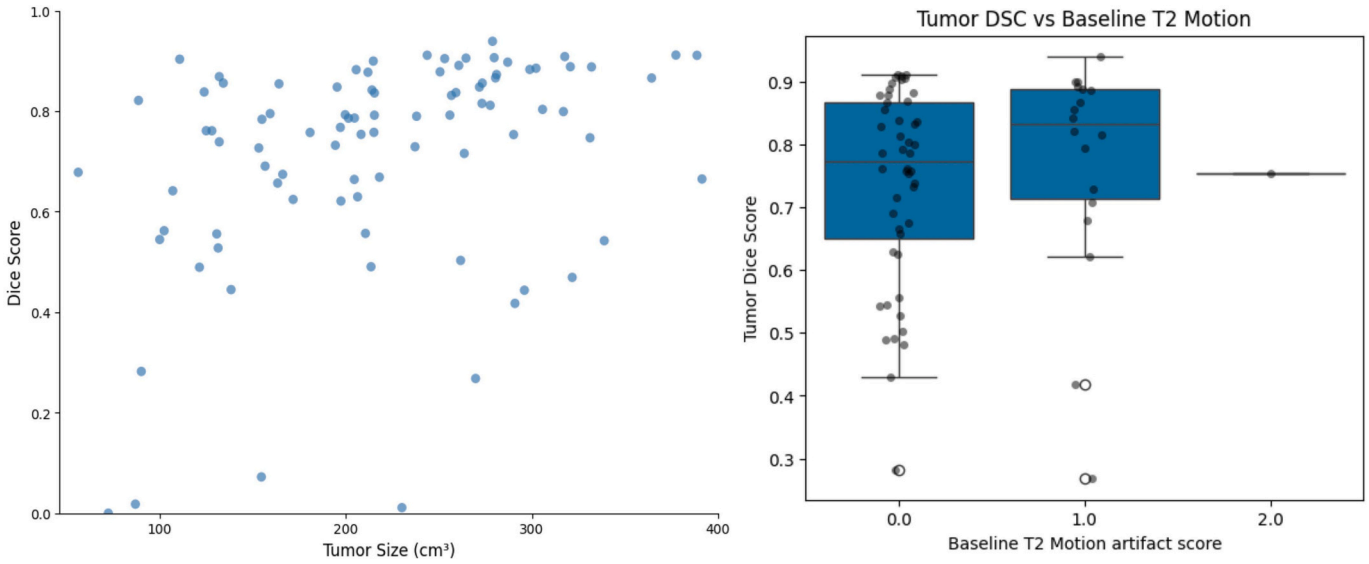


Fig. 5. Relationship between tumor size (cm^3) vs DSC (left) and Motion artifact scoring vs DSC (right). Tumor size was positively associated with segmentation accuracy for tumors with standard MRI presentation.

treatment is their use in developing daily online dose optimization strategies. This application of DL models has the potential to eliminate the need for internal target volume expansions, with the potential to reduce toxicity events related to intensity modulated radiation therapy [18].

External validation on the CCTH cohort provided critical insights into the generalizability of PocketNet. Despite differences in imaging protocols, institutions, and patient populations, the model maintained high segmentation accuracy, achieving mean DSC values of 80.8% for gynecologic organ segmentation and 67.3% for cervical tumor segmentation. This external validation confirmed that the model can accurately segment cervical, uterine, and vaginal tissues, as well as tumors, on T2-weighted MRI. These findings underscore the need for further model refinement, including the potential integration of diffusion-weighted imaging (DWI) to identify regions of restricted diffusion specific to cervical tumors, which has been shown to improve gross tumor volume delineation accuracy compared to T2-weighted imaging alone, particularly for less experienced clinicians [19].

This study had several limitations, including the size of this dataset and the lack of multi-center training data. The two-stage pipeline introduces a risk of error propagation, whereby inaccuracies in first-stage organ segmentation constrain second-stage tumor predictions. Segmentation performance was reduced for small tumors (< 3 cm) where T2-weighted signal differentiation from normal cervical stroma was limited, and for cases with motion artifacts. The model was trained on T2-weighted imaging alone; integration of complementary sequences such as DWI may improve robustness. By incorporating a more diverse dataset, the model can be trained on a broader range of tumor sizes, morphologies, and degrees of soft tissue contrast, leading to more consistent and accurate segmentations. In addition, ensuring that these diverse datasets are collected using consistent, standardized imaging protocols will minimize variability and improve the model.

In conclusion, a lightweight deep learning model achieved reliable automated segmentation of gynecologic organs and cervical tumors on T2-weighted MRI, with performance statistically equivalent to a full-sized nnUNet while requiring substantially less memory and training time. Successful external validation across a multi-center, multi-vendor dataset supports the generalizability of this approach. Further work will focus on expanding training diversity, integrating complementary imaging sequences, and prospective clinical validation, with the goal of supporting routine automated segmentation in cervical cancer radiotherapy planning.

CRediT authorship contribution statement

Awj Twam: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis. **Megan C. Jacobsen:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Data curation. **Adrian E. Celaya:** Software, Formal analysis. **Rachel Glenn:** Data curation. **Peng Wei:** Writing – review & editing, Validation, Formal analysis. **Jia Sun:** Writing – review & editing, Validation, Formal analysis. **Lilie Lin:** Data curation, Funding acquisition, Resources. **Ann Klopp:** Writing – review & editing, Supervision, Resources, Conceptualization. **Aradhana M. Venkatesan:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **David Fuentes:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Aradhana Venkatesan reports grant support from Siemens Healthineers, the Ingram Family Fund, the US Department of Defense, and the Joint Center for Computational Oncology Oncological Data and Computational Sciences Grant program, co-sponsored by the University of Texas MD Anderson Cancer Center, the Oden Institute for Computational Engineering and Sciences, and the Texas Advanced Computing Center (TACC). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported through the QIAC Partnership in Research (QPR) Program, Multidivisional Shark Tank, and the MD Anderson Strategic Research Initiative Development (STRIDE) Program – Tumor Measurement Initiative. Support from NIH R01CA195524 and NSF Award NSF-2111147 is gratefully acknowledged.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.phro.2026.101049.

Data availability

The MR images from MD Anderson are not publicly available at this time.

References

[1] Ferlay J, Colombet M, Soerjomataram I, Parkin DM, Piñeros M, Znaor A, et al. Cancer statistics for the year 2020: an overview. Int J Cancer 2021;149:778–9. https://doi.org/10.1002/ijc.33588.
[2] Burd EM. Human papillomavirus and cervical cancer. Clin Microbiol Rev 2003;16:1–17. https://doi.org/10.1128/CMR.16.1.1-17.2003.
[3] Singh D, Vignat J, Lorenzoni V, Eslahi M, Ginsburg O, Lauby-Secretan B, et al. Global estimates of incidence and mortality of cervical cancer in 2020: a baseline analysis of the WHO global cervical Cancer elimination initiative. Lancet Glob Health 2023;11:e197–206. https://doi.org/10.1016/S2214-109X(22)00501-0.
[4] Fischerova D, Frühauf F, Burgetova A, Haldorsen IS, Gatti E, Cibula D. The role of imaging in cervical cancer staging: ESGO/ESTRO/ESP guidelines (update 2023). Cancers 2024;16:775. https://doi.org/10.3390/cancers16040775.
[5] Zabihollahy F, Viswanathan AN, Schmidt EJ, Lee J. Fully automated segmentation of clinical target volume in cervical cancer from magnetic resonance imaging with convolutional neural network. J Appl Clin Med Phys 2022;23:e13725. https://doi.org/10.1002/acm2.13725.
[6] Tanderup K, Georg D, Pötter R, Kirisits C, Grau C, Lindegaard JC. Adaptive management of cervical cancer radiotherapy. Semin Radiat Oncol 2010;20:121–9. https://doi.org/10.1016/j.semradonc.2009.11.006.
[7] Fokdal L, Sturdza A, Mazeron R, Haie-Meder C, Tan LT, Gillham C, et al. Image guided adaptive brachytherapy with combined intracavitary and interstitial technique improves the therapeutic ratio in locally advanced cervical cancer. Radiother Oncol 2016;120:441–6. https://doi.org/10.1016/j.radonc.2016.05.014.
[8] Urushibara A, Saida T, Mori K, Ishiguro T, Sakai M, Masuoka S, et al. Diagnosing uterine cervical cancer on a single T2-weighted image: comparison between deep learning versus radiologists. Eur J Radiol 2021;135:109471. https://doi.org/10.1016/j.ejrad.2020.109471.
[9] Ma CY, Zhou JY, Xu XT, Guo J, Han MF, Gao YZ, et al. Deep learning-based auto-segmentation of clinical target volumes for radiotherapy treatment of cervical cancer. J Appl Clin Med Phys 2022;23:e13470. https://doi.org/10.1002/acm2.13470.
[10] Liu Z, Liu X, Guan H, Zhen H, Sun Y, Chen Q, et al. Development and validation of a deep learning algorithm for auto-delineation of clinical target volume and organs at risk in cervical cancer radiotherapy. Radiother Oncol 2020;153:172–9. https://doi.org/10.1016/j.radonc.2020.09.060.
[11] Isensee F, Jaeger PF, Kohl SA, Petersen J, Maier-Hein KH. nnU-net: a self-configuring method for deep learning-based biomedical image segmentation. Nat Methods 2021;18:203–11. https://doi.org/10.1038/s41592-020-01008-z.
[12] Celaya A, Actor JA, Muthusivarajan R, Gates E, Chung C, Schellingerhout D, et al. PocketNet: a smaller neural network for medical image analysis. IEEE Trans Med Imaging 2023;42:1172–84. https://doi.org/10.1109/TMI.2022.3224873.
[13] Mayr N, Yuh WTC, Bowen S, Harkenrider M, Knopp MV, Lee EYP, et al. Cervical cancer – tumor heterogeneity: serial functional and molecular imaging across the radiation therapy course in advanced cervical cancer [dataset]. Cancer Imaging Arch 2023. https://doi.org/10.7937/ERZ5-QZ59.
[14] Chung SY, Chang JS, Kim YB. Comprehensive clinical evaluation of deep learning-based auto-segmentation for radiotherapy in patients with cervical cancer. Front Oncol 2023;13:1119008. https://doi.org/10.3389/fonc.2023.1119008.
[15] Kano Y, Ikushima H, Sasaki M, Haga A. Automatic contour segmentation of cervical cancer using artificial intelligence. J Radiat Res 2021;62:934–44. https://doi.org/10.1093/jrr/rrab070.

[16] Hodneland E, Kaliyugarasan S, Wagner-Larsen KS, Lura N, Andersen E, Bartsch H, et al. Fully automatic whole-volume tumor segmentation in cervical cancer. Cancers (Basel) 2022;14:2372. https://doi.org/10.3390/cancers14102372.
[17] Lin YC, Lin CH, Lu HY, et al. Deep learning for fully automated tumor segmentation and extraction of magnetic resonance radiomics features in cervical cancer. Eur Radiol 2020;30:1297–305. https://doi.org/10.1007/s00330-019-06467-3.
[18] Rigaud B, Anderson BM, Yu ZH, Gobeli M, Cazoulat G, Söderberg J, et al. Automatic segmentation using deep learning to enable online dose optimization during adaptive radiation therapy of cervical cancer. Int J Radiat Oncol Biol Phys 2021;109:1096–110. https://doi.org/10.1016/j.ijrobp.2020.10.038.
[19] Zimmermann L, Knäusl B, Knoth J, Sturdza A, Dick V, Mrva-Ghukasyan T, et al. Diffusion weighted imaging for gross tumor volume delineation in primary radiochemotherapy and image guided adaptive brachytherapy for cervical cancer. Radiother Oncol 2026;214:111306. https://doi.org/10.1016/j.radonc.2025.111306.